VIVEKANAND EDUCATION SOCIETY
INSTITUTE OF TECHNOLOGY
(AUTONOMOUS)

SIMULATION LAB FOR SIGNAL PROCESSING

Experiment - 09

Auto and Cross Correlation on Continuous and Discrete Time Signals

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Aim: To study and verify the Auto Correlation Function (ACF) and Cross Correlation Function (CCF) and understand their properties for periodic, aperiodic, continuous-time, and discrete-time signals.

Software Used: MATLAB Software.

Theory: Correlation is the measure of similarity between two signals. There are two types:
Auto Correlation Function (ACF) measures the similarity between a signal x(t) and its time delayed version x(t+τ). It is represented as Rxx(τ) and is given by:

- $Rxx(\tau) = \int x(t)x(t+\tau)\, dt$ — for aperiodic signals
- $Rxx(\tau) = \lim 1/2T \int x(t)x(t+\tau)\, dt$ — for periodic signals

Cross Correlation Function (CCF) measures the similarity between two different signals x(t) and y(t). It is represented as Rxy(τ) and is given by:

- $Rxy(\tau) = \int x(t)y(t+\tau)\, dt$ — for aperiodic signals
- $Rxy(\tau) = \lim 1/2T \int x(t)y(t+\tau)\, dt$ — for periodic signals.

Procedure:

1. Open MATLAB and create a new script file in the editor .
2. Define the time vector and the signals x(t) and y(t) using suitable mathematical expressions.
3. Compute the Auto Correlation of x(t) using the MATLAB command **xcorr ()**.
4. Compute the Cross Correlation between x(t) and y(t) using **xcorr ().**
5. Plot the original signals x(t) and y(t) using **plot ()** for continuous-time and **stem ()** for discrete-time signals for visualization.
6. Plot the ACF and CCF results using **subplot ()** and **plot ()** for comparison.
7. Observe the shape, symmetry, and peak values of the correlation functions.
8. Analyze the output plots and record your observations.

Codes:

1. **To perform Cross Correlation on two user-input discrete-time sequences x(n) and h(n) using "xcorr ()":**

⇨
```
x = [1, 2, 3, 4, 5];
h = [1, 1, 1, 1];
y = xcorr (x, h);
disp ('The resultant signal is:');
disp (y);
figure;
subplot (3,1,1);
stem (0: length(x)-1, x, 'b', 'filled');
xlabel('n');
ylabel('x(n)');
title ('Input Signal x(n)');
grid on;
subplot (3,1,2);
stem (0: length(h)-1, h, 'rs', 'filled');
xlabel('n');
ylabel('h(n)');
title ('Impulse Signal h(n)');
grid on;
subplot (3,1,3);
stem (0: length(y)-1, y, 'g^', 'filled');
xlabel('n');
ylabel('y(n)');
title ('Correlation Signal y(n) = xcorr (x, h)');
grid on;
sgtitle ('Cross Correlation');
```

2. **To perform Auto Correlation on discrete-time sequences x(n) = [1,2,3,4,5] and y(n) = [4,1,5,2,6] using "xcorr ()":**

⇨
```
x = [1,2,3,4,5];
y = [4,1,5,2,6];
z = xcorr(x,x);
disp ('Auto-correlation result z:');
disp(z);
figure;
subplot (3,1,1);
stem (0: length(x)-1, x, 'b', 'filled');
xlabel('n');
ylabel('x(n)');
title ('Input Signal x(n)');
grid on;
subplot (3,1,2);
stem (0: length(y)-1, y, 'rs', 'filled');
```

```
xlabel('n');
ylabel('y(n)');
title ('Input Signal y(n)');
grid on;
subplot (3,1,3);
stem (0: length(z)-1, z, 'g^', 'filled');
xlabel('n');
ylabel('z(n)');
title ('Resultant Signal z(n) = xcorr (x, x)');
grid on;
sgtitle ('Auto Correlation');
```

3. To perform Cross Correlation on continuous-time sinusoidal signals $x(t) = \sin(2\pi f_1 t)$ and $y(t) = \sin(2\pi f_2 t)$ using the manual method:

⇨
```
f1 = 100;
T1 = 1/f1;
t1 = 0: T1/f1: T1;
x = sin(2*pi*f1*t1);
f2 = 150;
T2 = 1/f2;
t2 = 0: T2/f2: T2;
y = sin(2*pi*f2*t2);
t22 = -fliplr(t2);
y1 = fliplr(y);
A = zeros(min(length(x), length(y1)), length(x)+length(y1)-1);
if length(x) < length(y1)
   for i = 1: length(x)
      A (i, i: i+length(y1)-1) = y1(1, :);
      A (i, :) = x(i). * A (i, :);
   end
else
   for i = 1: length(y1)
      A (i, i: i+length(x)-1) = x (1, :);
      A (i, :) = y1(i). * A (i, :);
   end
end
c = zeros (1, length(x)+length(y1)-1);
for i = 1: min(length(x), length(y1))
   c (1, :) = c (1, :) + A (i, :);
end
t = linspace(min(t1) +min(t22), max(t1) +max(t22), length(c));
figure;
subplot (3,1,1);
plot (t1, x, 'b', 'LineWidth', 1.5);
xlabel('Time');
ylabel('Amplitude');
```

```
title (['Given Signal x(t) with f = ' num2str(f1)]);
grid on;
subplot (3,1,2);
plot (t22, y1, 'r', 'LineWidth', 1.5);
xlabel('Time');
ylabel('Amplitude');
title (['Signal y(-t) with f = ' num2str(f2)]);
grid on;
subplot (3,1,3);
plot (t, c, 'g', 'LineWidth', 1.5);
xlabel('Time');
ylabel('Amplitude');
title ('Cross Correlation of x(t) and y(t)');
grid on;
sgtitle ('Cross Correlation — Continuous Time Signals');
```

4. To perform Cross Correlation on two user-input discrete-time sequences x(n) and y(n) with defined time indices using the manual flip-and-shift method:

⇨
```
x = [1 2 3 4 5];
n1 = [0 1 2 3 4];
y = [1 1 2 1];
n2 = [0 1 2 3];
n22 = -fliplr(n2);
y1 = fliplr(y);
n = min(n1) +min(n22): max(n1) +max(n22);
A = zeros(min(length(x), length(y1)), length(x)+length(y1)-1);
if length(x) < length(y1)
   for i = 1: length(x)
      A (i, i: i+length(y1)-1) = y1;
      A (i, :) = x(i). * A (i, :);
   end
else
   for i = 1: length(y1)
      A (i, i: i+length(x)-1) = x;
      A (i, :) = y1(i). * A (i, :);
   end
end
c = zeros (1, length(x)+length(y1)-1);
for i = 1: min(length(x), length(y1))
   c = c + A (i, :);
end
disp ('The Cross correlation between the given sequences is:');
disp(c);
disp ('The Time index of the Cross correlation is:');
disp (n);
figure;
```

```
subplot (3,1,1);
stem (n1, x, 'b', 'filled');
xlabel('Time');
ylabel('Amplitude');
title ('Given Sequence x(n)');
grid on;
subplot (3,1,2);
stem (n22, y1, 'rs', 'filled');
xlabel('Time');
ylabel('Amplitude');
title ('Sequence y(-n)');
grid on;
subplot (3,1,3);
stem (n, c, 'g^', 'filled');
xlabel('Time');
ylabel('Amplitude');
title ('Cross Correlation of x(n) and y(n)');
grid on;
sgtitle ('Cross Correlation — Discrete Time Sequences');
```

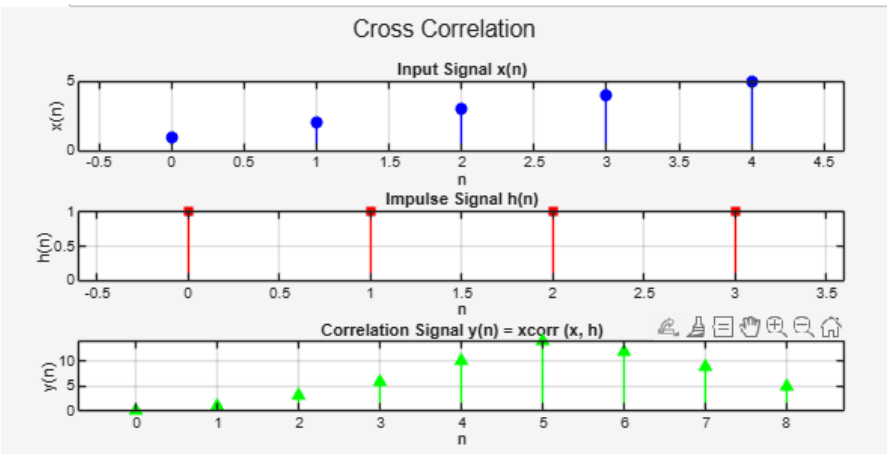
MATLAB Code Snippets:

Q1:

Expt9Eklavya.m

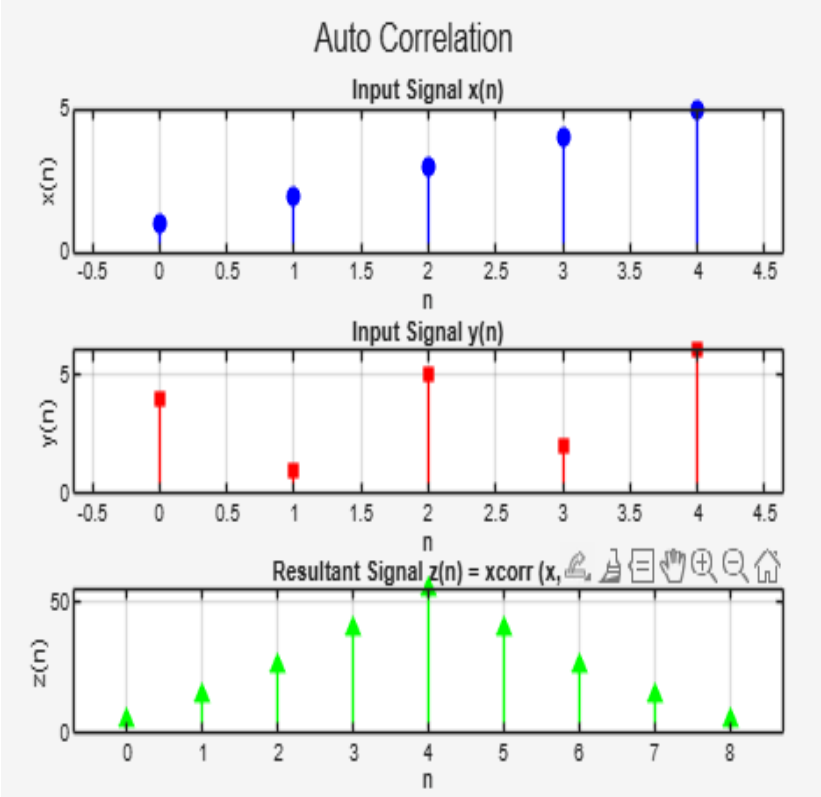
MATLAB Drive/Expt9Eklavya.m

```
x = [1, 2, 3, 4, 5];
h = [1, 1, 1, 1];
y = xcorr (x, h);
disp ('The resultant signal is:');
disp (y);
figure;
subplot (3,1,1);
stem (0: length(x)-1, x, 'b', 'filled');
xlabel('n');
ylabel('x(n)');
title ('Input Signal x(n)');
grid on;
subplot (3,1,2);
stem (0: length(h)-1, h, 'rs', 'filled');
xlabel('n');
ylabel('h(n)');
title ('Impulse Signal h(n)');
grid on;
subplot (3,1,3);
stem (0: length(y)-1, y, 'g^', 'filled');
xlabel('n');
ylabel('y(n)');
title ('Correlation Signal y(n) = xcorr (x, h)');
grid on;
sgtitle ('Cross Correlation');
```



Q2:

```
x = [1,2,3,4,5];
y = [4,1,5,2,6];
z = xcorr(x,x);
disp ('Auto-correlation result z:');
disp(z);
figure;
subplot (3,1,1);
stem (0: length(x)-1, x, 'b', 'filled');
xlabel('n');
ylabel('x(n)');
title ('Input Signal x(n)');
grid on;
subplot (3,1,2);
stem (0: length(y)-1, y, 'rs', 'filled');
xlabel('n');
ylabel('y(n)');
title ('Input Signal y(n)');
grid on;
subplot (3,1,3);
stem (0: length(z)-1, z, 'g^', 'filled');
xlabel('n');
ylabel('z(n)');
title ('Resultant Signal z(n) = xcorr (x, x)');
grid on;
sgtitle ('Auto Correlation');
```

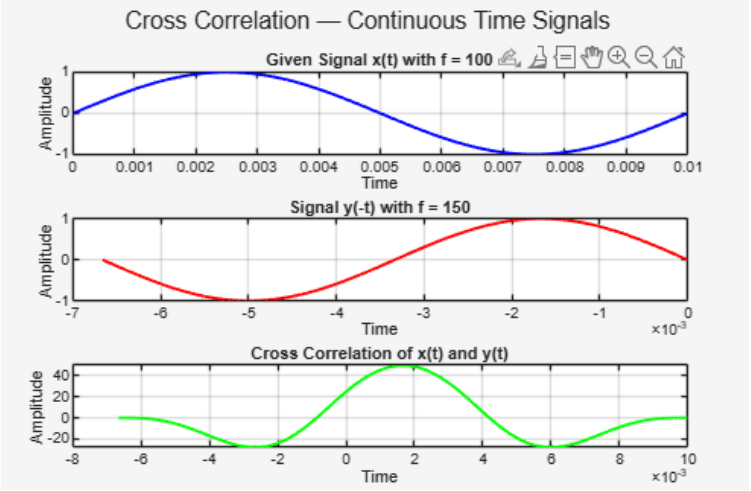



```
>> Expt9Eklavya
Auto-correlation result z:
    5.0000   14.0000   26.0000   40.0000   55.0000   40.0000   26.0000   14.0000    5.0000
```

Q3:

MATLAB Drive/Expt9Eklavya.m

```
f1 = 100;
T1 = 1/f1;
t1 = 0: T1/f1: T1;
x = sin(2*pi*f1*t1);
f2 = 150;
T2 = 1/f2;
t2 = 0: T2/f2: T2;
y = sin(2*pi*f2*t2);
t22 = -fliplr(t2);
y1 = fliplr(y);
A = zeros(min(length(x), length(y1)), length(x)+length(y1)-1);
if length(x) < length(y1)
    for i = 1: length(x)
        A (i, i: i+length(y1)-1) = y1(1, :);
        A (i, :) = x(i).* A (i, :);
    end
else
    for i = 1: length(y1)
        A (i, i: i+length(x)-1) = x (1, :);
        A (i, :) = y1(i).* A (i, :);
    end
end
c = zeros (1, length(x)+length(y1)-1);
for i = 1: min(length(x), length(y1))
    c (1, :) = c (1, :) + A (i, :);
end
t = linspace(min(t1) +min(t22), max(t1) +max(t22), length(c));
figure;
subplot (3,1,1);
plot (t1, x, 'b', 'LineWidth', 1.5);
xlabel('Time');
ylabel('Amplitude');
title (['Given Signal x(t) with f = ' num2str(f1)]);
grid on;
subplot (3,1,2);
plot (t22, y1, 'r', 'LineWidth', 1.5);
xlabel('Time');
ylabel('Amplitude');
title (['Signal y(-t) with f = ' num2str(f2)]);
grid on;
subplot (3,1,3);
plot (t, c, 'g', 'LineWidth', 1.5);
xlabel('Time');
ylabel('Amplitude');
title ('Cross Correlation of x(t) and y(t)');
grid on;
sgtitle ('Cross Correlation — Continuous Time Signals');
```

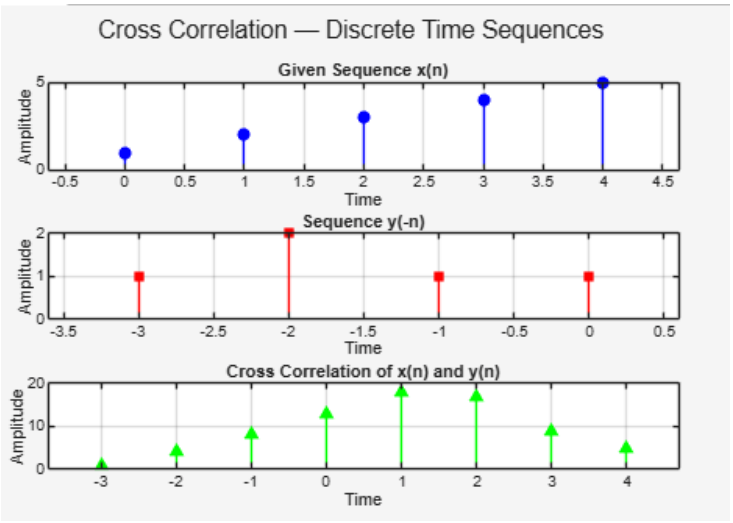


Q4:

Expt9Eklavya.m

MATLAB Drive/Expt9Eklavya.m

```
x = [1 2 3 4 5];
n1 = [0 1 2 3 4];
y = [1 1 2 1];
n2 = [0 1 2 3];
n22 = -fliplr(n2);
y1 = fliplr(y);
n = min(n1) +min(n22): max(n1) +max(n22);
A = zeros(min(length(x), length(y1)), length(x)+length(y1)-1);
if length(x) < length(y1)
    for i = 1: length(x)
        A (i, i: i+length(y1)-1) = y1;
        A (i, :) = x(i).* A (i, :);
    end
else
    for i = 1: length(y1)
        A (i, i: i+length(x)-1) = x;
        A (i, :) = y1(i).* A (i, :);
    end
end
c = zeros (1, length(x)+length(y1)-1);
for i = 1: min(length(x), length(y1))
    c = c + A (i, :);
end
disp ('The Cross correlation between the given sequences is:');
disp(c);
disp ('The Time index of the Cross correlation is:');
disp (n);
figure;
subplot (3,1,1);
stem (n1, x, 'b', 'filled');
xlabel('Time');
ylabel('Amplitude');
title ('Given Sequence x(n)');
grid on;
subplot (3,1,2);
stem (n22, y1, 'rs', 'filled');
xlabel('Time');
ylabel('Amplitude');
title ('Sequence y(-n)');
grid on;
subplot (3,1,3);
stem (n, c, 'g^', 'filled');
xlabel('Time');
ylabel('Amplitude');
title ('Cross Correlation of x(n) and y(n)');
grid on;
sgtitle ('Cross Correlation — Discrete Time Sequences');
```




```
>> Expt9Eklavya
The Cross correlation between the given sequences is:
     1     4     8    13    18    17     9     5

The Time index of the Cross correlation is:
    -3    -2    -1     0     1     2     3     4
```

Conclusion:

In this experiment, the Auto Correlation Function (ACF) and Cross Correlation Function (CCF) were studied and verified using MATLAB for both continuous and discrete-time signals. It was observed that the ACF is always maximum at zero shift, indicating maximum self-similarity. The CCF measured the similarity between two different signals and was found to be non-symmetric unlike ACF. For discrete-time sequences, correlation was computed using both the "**xcorr ()**" function and the manual flip-and-shift method, with both yielding consistent results. Thus, the experiment successfully verifies the properties of ACF and CCF for continuous and discrete-time signals.